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VEHICLE INCREMENTAL MOVEMENT SYSTEM

Abstract

A system includes a computer having a processor and a memory storing instructions executable by the processor to: receive at least one discrete input within a predetermined time period from a human-machine interface while the vehicle is at a standstill; and in response to the at least one discrete input, move the vehicle an incremental distance and returning the vehicle to a standstill.

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Background/Summary

BACKGROUND

[0001] During certain vehicle maneuvers, a human driver may seek to stop movement of the vehicle, make a small movement, and return the vehicle to a stop. In order to perform such a maneuver, the human driver typically remove a foot from a brake pedal, taps the accelerator pedal, and again steps on the brake pedal while monitoring the position of the vehicle. One such example is a parallel parking maneuver in which the human driver uses the brake, acceleration, brake sequence in both forward vehicle movement and rearward vehicle movement to position the vehicle between two parked vehicles. Another example is during alignment of a hitch of the vehicle with a trailer in which the vehicle is typically moving rearward. Other examples may be during operation of the vehicle off-road including operations such as maneuvering on or around obstacles, e.g., ground obstacles like rocks, tree stumps, etc. In some examples, the human driver may also experience and account for creep torque during such maneuvers. When the vehicle experiences creep torque, the vehicle can drive at a slow speed when no pressure is applied to the accelerator pedal or to the brake pedal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0002] FIG. 1A is an example vehicle including an incremental-movement system, wherein the vehicle has moved an incremental distance by operation of the incremental-movement system.

[0003] FIG. 1B is the example vehicle of FIG. 1A, wherein the vehicle has moved an incremental distance different than that in FIG. 1A by operation of the incremental-movement system.

[0004] FIG. 2 is a plan view of a vehicle in the process of a parallel parking maneuver with the use of the incremental-movement system.

[0005] FIG. 3 is a schematic view of the incremental-movement system.

[0006] FIG. 4 is an example method performed by the incremental-movement system.

[0007] FIG. 5A includes example graphs of one example of the incremental-movement system showing entry of one discrete input to the incremental-movement system, the corresponding velocity of the vehicle under operation of the incremental-movement system, and distance traveled by the vehicle under operation of the incremental-movement system.

[0008] FIG. 5B includes example graphs of the example incremental-movement system of FIG. 5A showing entry of three discrete inputs to the incremental-movement system in a time period, the corresponding velocity of the vehicle under operation of the incremental-movement system, and distance traveled by the vehicle under operation of the incremental-movement system.

[0009] FIG. 6 includes example graphs of another example of the incremental-movement system showing entry of three discrete inputs to the incremental-movement system in a time period, the corresponding velocity of the vehicle under operation of the incremental-movement system, and distance traveled by the vehicle under operation of the incremental-movement system.

[0010] FIG. 7 includes example graphs of another example incremental-movement system showing entry of three discrete inputs to the incremental-movement system in a time period, the corresponding velocity of the vehicle under operation of the incremental-movement system, and distance traveled by the vehicle under operation of the incremental-movement system.

DETAILED DESCRIPTION

[0011] This disclosure describes a system and method for controlling incremental movements of a vehicle under certain conditions when one or more discrete inputs are entered by a human driver of the vehicle in a time period. The conditions include that the vehicle is at a standstill vehicle when the one or more discrete inputs are entered by the human driver. When the conditions are met and the one or more discrete inputs are entered by the human driver, an incremental movement of the vehicle starts when the vehicle is at standstill and ends when the vehicle returns to a standstill. The

discrete inputs are entered by the human driver through an incremental-movement user interface, e.g., a human-machine interface (HMI) such as a button, paddle, touchscreen, etc. In some examples, the incremental-movement user interface is on a steering wheel of the vehicle. The incremental-movement user interface is configured so that the discrete inputs are discrete from each other, i.e., individual and distinct from each other. In some examples, a single finger depression and finger release of the incremental-movement user interface (e.g., depressing and releasing a button with a finger, depressing and releasing a paddle with a finger, touching and untouching an icon on a touchscreen) corresponds to one discrete input. In examples in which more than one discrete input is entered through the incremental-movement user interface in a time period, the discrete inputs are individual and distinct from each other. For example, one finger depression and finger release is one discrete input and another finger depression and finger release in the time period is a second discrete input.

[0012] After entry of the one or more discrete inputs, e.g., at the end of the time period, the system moves the vehicle an incremental distance. The length of the incremental distance is based on the number of discrete inputs entered through the incremental-movement user interface in the time period. The length of the incremental distance is increased by additional discrete inputs entered through the incremental-movement user interface in the time period. In some examples, components of the vehicle are controlled by a closed-loop system to incrementally move the vehicle, and in other examples, components of the vehicle are controlled by an open-loop system to incrementally move the vehicle, as described further below.

[0013] Multiple, separate, incremental movements of the vehicle may be controlled by the human driver by discrete inputs in different time periods. In such instances, the incremental movements can be consecutively applied by the human driver and/or can be interspersed with control of the movement of the vehicle by other ways of controls a powertrain system of the vehicle, e.g. control with an accelerator pedal and/or brake pedal. As examples, the human driver may operate movement of the vehicle with multiple incremental movements, whether consecutive or interspersed with other movement control, to position the vehicle relative to stationary objects. For example, the human driver may operate the vehicle with one or more incremental movements to parallel park the vehicle. As another example, the human driver may operate the vehicle with one or more incremental movements to position a hitch of the vehicle relative to a trailer to be connected to the hitch. As another example, the human driver may operate the vehicle with one or more incremental movements to position a wheel of a vehicle next to or on top of a ground feature, e.g. a curb, a rock or stump during off-road use, etc. The human driver can steer the vehicle before and during the incremental movement of the vehicle to position the vehicle in the desired location relative to an exterior object. Operations, systems, and methods described herein should always be implemented and/or performed in accordance with an applicable owner's/user's manual and/or safety guidelines.

[0014] A system includes a computer having a processor and a memory storing instructions executable by the processor to: receive at least one discrete input within a predetermined time period from a human-machine interface while the vehicle is at a standstill; and in response to the at least one discrete input, move the vehicle an incremental distance and returning the vehicle to a standstill.

[0015] In some examples, the memory stores instructions executable by the processor to: receive an indication from a steering angle sensor that a steering angle of a steering wheel is outside of a predetermined range; and move the vehicle the incremental distance based on the indication that the steering angle is outside of the predetermined range. In such examples, the system may include the steering wheel of the vehicle and the steering angle sensor;

[0016] In some examples, the memory stores instructions executable by the processor to command a traction motor to move the vehicle in the incremental distance.

[0017] In some examples, the memory stores instructions executable by the processor to brake the

wheel of the vehicle during movement of the vehicle in the incremental distance to return the vehicle to a standstill. For example, the system may include a wheel powered by the traction motor and a brake on the wheel; and the memory may store instructions executable by the processor to brake the wheel of the vehicle during movement of the vehicle in the incremental distance to return the vehicle to a standstill.

[0018] In some examples, the memory stores instructions executable by the processor to: receive more than one discrete input within the predetermined time period; and calculate the length of the incremental distance based on the number of discrete inputs. For example, the system may include a traction motor, and the memory may store instructions executable by the processor to: determine a motor input for at least one traction motor based on the length of the incremental distance; and command the at least one traction motor to rotate based on the motor input.

[0019] In some examples, the memory may store instructions executable by the processor to: receive more than one discrete input within the predetermined time period; and access a lookup table to identify a length of the incremental distance based on the number of discrete inputs.

[0020] In some examples, the memory may store instructions executable by the processor to: command a first traction motor to rotate a first wheel while measuring travel of a second wheel; then, identify occurrence of gear lash in the rotation of the second wheel; and then, command a second traction motor to rotate the second wheel while measuring travel of the first wheel. In such examples, the system may include the first traction motor, the second traction motor, the first wheel powered by the first traction motor, and the second wheel powered by the second traction motor.

[0021] In some examples, the memory may store instructions executable by the processor to: command the traction motor to initiate movement in the incremental step from standstill; measure travel of the wheel of the vehicle from standstill; and return the vehicle to standstill based on measurement of the travel of the wheel. In such examples, the system may include a traction motor and a wheel powered by the traction motor.

[0022] With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a computer (hereinafter referred to as “vehicle computer **105**”) includes a processor and a memory, and the memory stores instructions executable by the processor to receive at least one discrete input within a predetermined time period from an incremental-movement user interface **115** while the vehicle **100** is at a standstill; and in response to the at least one discrete input, move the vehicle **100** an incremental distance (identified as “d” in FIGS. 1A-B) and returning the vehicle **100** to a standstill.

[0023] With reference to FIGS. 1A-2, the vehicle **100** may be any passenger or commercial automobile such as a car, a truck, a sport utility vehicle, a crossover, a van, a minivan, a taxi, a bus, etc.

[0024] With reference to FIG. 3, a system (hereinafter the “incremental-movement system **160**”) includes the vehicle computer **105** is a microprocessor-based computing device, e.g., a generic computing device including a processor and a memory, an electronic controller or the like, a field-programmable gate array (FPGA), an application-specific integrated circuit (ASIC), a combination of the foregoing, etc. Typically, a hardware description language such as VHDL (VHSIC (Very High Speed Integrated Circuit) Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected to the FPGA circuit. The vehicle computer **105** can thus include a processor, a memory, etc. The memory of the vehicle computer **105** can include media for storing instructions executable by the processor as well as for electronically storing data and/or databases, and/or the vehicle computer **105** can include structures such as the foregoing by which programming is provided. The vehicle computer **105** can be multiple computers coupled together.

[0025] The vehicle computer **105** may transmit and receive data through a communications

network **155** such as a controller area network (CAN) bus, Ethernet, WiFi, Local Interconnect Network (LIN), onboard diagnostics connector (OBD-II), and/or by any other wired or wireless communications network. The vehicle computer **105** may be communicatively coupled to a propulsion system **150**, a brake system **120**, a steering system **125**, sensors, an accelerator pedal, a brake pedal, the incremental-movement user interface **115**, and other components via the communications network **155**.

[0026] The propulsion system **150** of the vehicle **100** generates energy and translates the energy into motion of the vehicle **100**. The propulsion system **150** may be a conventional vehicle propulsion system, for example, a conventional powertrain including an internal-combustion engine coupled to a transmission that transfers rotational motion to wheels **145**; an electric powertrain including batteries and one or more electric motors, i.e., a traction motor **110**, that transfer rotational motion to wheels **145** of the vehicle **100**; a hybrid powertrain including elements of the conventional powertrain and the electric powertrain; or any other type of propulsion. The propulsion system **150** can include an electronic control unit (ECU) or the like, e.g., a powertrain control module, that is in communication with and receives input from the computer and/or a human driver. The human driver may control the propulsion system **150** via, e.g., the accelerator pedal and/or a gear-shift lever, as well as with the incremental-movement user interface **115** as described below.

[0027] In examples in which the propulsion system **150** is an electric powertrain or a hybrid powertrain, the propulsion system **150** may include a traction motor **110**. In such an example, each traction motor **110** transfers rotational motion to the respective wheel **145** based on input from the communications network **155**, e.g. from the powertrain control module. In some examples, the traction motors **110** may be a known type. For example, the traction motor **110** may be of a known type for propulsion of the vehicle **100** in an electric or hybrid vehicle. The traction motors **110** may be independently operable by the vehicle computer **105**. In other words, each traction motors **110** can receive its own input from the vehicle computer **105** and the input from the vehicle computer **105** to the traction motors **110** may be different for any one or all of the traction motors **110**. The wheel **145** transmits rotation from the traction motor **110** to the ground to propel the vehicle **100**. The wheel **145** may include a rim and a tire, as is known.

[0028] In some examples, the traction motor **110** may drive an axle. In other examples, the traction motor **110** may be a wheel hub motor that drives an individual wheel **145**. In some examples, the propulsion system **150** may include one, single traction motor **110**, e.g., to drive a front axle or a rear axle. As another example, the propulsion system **150** may include two traction motors **110**, e.g., one traction motor **110** to drive a front axle and one traction motor **110** to drive a rear axle. As another example, the propulsion system **150** may include three traction motors **110**, e.g., one traction motor **110** driving a front axle and two traction motors **110** driving two rear wheels **145**, respectively. As another example, the propulsion system **150** may include four traction motors **110**, e.g., one traction motor **110** at each wheel **145**, respectively.

[0029] As described herein, the propulsion system **150** can be controlled, in part, by operation of the incremental-movement user interface **115** under certain conditions. In addition to the incremental-movement user interface **115**, the propulsion system **150** may include an accelerator pedal to control the operation of the propulsion system **150**. The accelerator pedal provides input to the computer indicating a position of the accelerator pedal for use in controlling the propulsion system **150** and/or the brake system **120**. The accelerator pedal is positioned to be pressable by an operator of the vehicle **100**. The accelerator pedal is movable relative to the body of the vehicle **100**, e.g., is coupled by a hinge to the body of the vehicle **100**. The accelerator has a range of motion, e.g., an angular range of motion around the hinge. The position of the accelerator pedal within the range of motion is communicated to the vehicle computer **105**. In some examples, the accelerator pedal is at a floor of the vehicle **100** for control by a foot of a human driver. In other examples, the accelerator pedal may be in other locations in the interior of the vehicle **100**, e.g., a

paddle on the steering wheel **145**.

[0030] The brake system **120** is typically a conventional vehicle braking subsystem and resists the motion of the vehicle **100** to slow and/or stop the vehicle **100**. The braking system slows or stops rotation of the wheels **145** relative to the ground. In some examples, the brake system **120** may engage components of the wheels **145** to stop the wheels **145**, e.g., may include friction brakes such as disc brakes, drum brakes, band brakes, etc. In addition or in the alternative, in some examples, the traction motor **110** may brake the respective wheel **145**. In such an example, component of the traction motor **110** slows or stops rotation of the respective wheel **145** relative to the ground. In some examples, the brakes may be regenerative brakes. The brakes may be of any suitable type of brakes, including, in some examples, those that are known.

[0031] The brake system **120** can include an electronic control unit (ECU) or the like, e.g., a brake control module, that is in communication with and receives input from the vehicle computer **105** and/or a human driver. The human driver may control the brake system **120** via, e.g., the brake pedal.

[0032] The brake system **120** may include brake pedal that provides input to the vehicle computer **105** indicating a position of the brake pedal for use in controlling the brake system **120**. The brake pedal is positioned to be pressable by an operator of the vehicle **100**. The brake pedal is movable relative to the body of the vehicle **100**, e.g., is coupled by a hinge to the body of the vehicle **100**. The brake has a range of motion, e.g., an angular range of motion around the hinge. The position of the brake pedal within the range of motion is reported to the computer. In some examples, the brake pedal is at a floor of the vehicle **100** for control by a foot of a human driver. In other examples, the brake pedal may be in other locations in the interior of the vehicle **100**, e.g., a paddle on the steering wheel **145**.

[0033] The steering system **125** controls the turning of the wheels **145** to steer the direction of travel of the vehicle **100**. The steering system **125** may be a rack-and-pinion system with electric power-assisted steering, a steer-by-wire system, as both are known, or any other suitable system. The steering system **125** can include an electronic control unit (ECU) or the like, e.g., a steering control module, that is in communication with and receives input from the computer and/or a human driver. The human driver may control the steering system **125** via, for example, a steering wheel **145**.

[0034] In some examples, the steering system **125** includes a steering-wheel position sensor **130**. The steering-wheel position sensor **130** measures the rotational position of the steering wheel **145** relative to the body of the vehicle **100**. Specifically, the steering-wheel position sensor **130** measures the position of the steering wheel **145** that steers travel of the vehicle **100**. As an example, the steering-wheel position sensor **130** may be a rotational sensor that senses the rotational position of the steering wheel **145**, e.g., by measurement of the steering wheel **145** or a steering shaft connected to the steering wheel **145**.

[0035] Measurements from the steering-wheel position sensor **130** may be used to identify whether the rotational position of the steering wheel **145** is within or outside of a predetermined range. As an example, an example coordinate system may label the centered position of the steering wheel **145** (i.e., the position of the steering wheel **145** when steering the vehicle **100** in a straight line) as 0 degrees. As one example, the predetermined range may be ± 30 degrees from center. The vehicle computer **105** may receive communication of the angular position of the steering wheel **145** from the steering-wheel position sensor **130** and may determine whether the angular position is within the predetermined range (e.g., less than ± 30 degrees from center) or outside of the predetermined range (e.g., greater than ± 30 degrees from center).

[0036] In some examples, the vehicle **100**, e.g., the propulsion system **150**, may include a wheel-position sensor **135** for each wheel **145**. The wheel-position sensor **135** senses the rotational position of the wheel **145**, e.g., relative to a body of the vehicle **100**. The wheel-position sensor **135** may be, for example, a rotary encoder. In some examples, the rotary encoder may be a component

of an anti-lock brake system. The vehicle **100** may include a wheel-position sensor **135** at each wheel **145**.

[0037] In some examples, the vehicle **100** may include a resolver **140** at each wheel **145**. The resolver **140** measures the rotational position of the wheel **145**, e.g., relative to a body of the vehicle **100**. The resolver **140** may measure the rotational position in degrees of rotation. The resolver **140** may be an analog device or a digital device. In some examples, the resolver **140** may be a component of the traction motor **110**.

[0038] The vehicle **100** may be operable in one-pedal drive mode. When the one-pedal drive mode is disengaged (or when a vehicle **100** is not equipped with one-pedal drive), the vehicle **100** can drive at a slow speed when no pressure is applied to the accelerator pedal or to the brake pedal. This is called creep torque. When in one-pedal drive mode, the vehicle **100** will accelerate when pressure is applied to the accelerator pedal, and the vehicle **100** will brake when no pressure is applied to the accelerator pedal, even if no pressure is applied to the brake pedal. In other words, the vehicle **100** does not use creep torque when one-pedal drive mode is engaged.

[0039] The vehicle **100** may be operable in an automatic vehicle hold mode. In response to the automatic vehicle hold being active, the computer brakes the vehicle **100** in response to a brake pedal of the vehicle **100** being released while the vehicle **100** is at a standstill. In other words, when the vehicle **100** is at a standstill, the vehicle **100** remains at a standstill until the computer receives an input to accelerate. Thus, the vehicle **100** does not exhibit creep torque when stopped.

[0040] The vehicle **100** may be operable in a low-speed mode. When the vehicle **100** is in the low-speed mode, the vehicle computer **105** is programmed to control acceleration and braking of the vehicle **100** based on input from the components of the propulsion system **150** and/or the braking system of the vehicle **100** according to the low-speed mode. In the low-speed mode, acceleration occurs upon receiving input from the human driver. The vehicle computer **105** actuates the propulsion system **150** to accelerate the vehicle **100** upon receiving input from the human driver, e.g., through the acceleration pedal. The acceleration may be capped at a maximum torque or maximum speed. The maximum torque or maximum speed may be chosen to keep the vehicle **100** within speeds appropriate for parking, trailering, or similar maneuvers. In the low-speed mode, braking occurs upon not receiving input from the accelerator pedal or from any of the alternative input devices. In other words, the vehicle computer **105** actuates the brake system **120** to slow or stop the vehicle **100** upon not receiving input from any of the input devices that are designated to provide input controlling acceleration, even if the brake pedal is not being pressed. The braking force applied by the brake system **120** may depend on circumstances such as how the low-speed mode was engaged. For example, the braking force may be applied by regenerative brakes of the brake system **120** and may slow the vehicle **100** more gradually (sometimes called one-pedal drive), or the braking force may be applied by the disc brakes of the brake system **120** at a typical braking force used when the brake pedal is engaged. In the low-speed mode, braking also occurs upon receiving an input from the brake pedal, in the same manner as occurs in a two-pedal-drive mode.

[0041] The vehicle computer **105** is programmed to, in response to the low-speed mode being inactive, control acceleration and braking of the vehicle **100** not according to the low-speed mode. For example, the vehicle **100** may be in the two-pedal-drive mode. In the two-pedal-drive mode, acceleration occurs upon receiving input from the accelerator pedal, and braking occurs upon receiving input from the brake pedal. If neither the accelerator pedal nor the brake pedal is providing input (i.e., are pressed by the operator), then friction will slow the vehicle **100**, and the vehicle **100** will exhibit creep torque at low speeds.

[0042] The vehicle computer **105** is programmed to switch the vehicle **100** between various modes, e.g., one-pedal drive mode, two-pedal drive mode, low-speed mode, and automatic vehicle hold mode, under various conditions. The method **400** is operable in any suitable vehicle mode as long as the vehicle **100** is at a standstill. The vehicle **100** being at a standstill is a condition for operation

of method **400**. The vehicle **100** is at a standstill when all wheels **145** of the vehicle **100**, i.e., all four wheels **145** in four-wheeled vehicle examples, are stationary relative to the body of the vehicle **100** and the ground.

[0043] The incremental-movement user interface **115** is a human-machine interface (HMI) designed to receive discrete inputs from the human driver. The incremental-movement user interface **115** may be, for example, a button, paddle, touchscreen, etc., designed receiving input from the human driver. The incremental-movement user interface **115** may be in the interior of the vehicle **100**. In other words, in such examples, the incremental-movement user interface **115** is in the passenger cabin. In such examples, the incremental-movement user interface **115** is accessible by the human driver of the vehicle **100** to operate the incremental-movement user interface **115** during operation of the vehicle **100**. In such examples, the incremental-movement user interface **115** is positioned in the passenger cabin so that the incremental-movement user interface **115** can be reached by the human operator of the vehicle **100** while the human operator remains seated and controlling the steering wheel **145**. In some examples, the incremental-movement user interface **115** is on the steering wheel **145** of the vehicle **100**. In other examples, the incremental-movement user interface **115** may be on an exterior of the vehicle.

[0044] The incremental-movement user interface **115** is configured so that the discrete inputs are discrete from each other, i.e., individual and distinct from each other. The incremental-movement user interface **115** has a binary operation, i.e., the incremental-movement user interface **115** has two positions, activated by the human driver or not activated by the human driver. In some examples, a single finger depression and finger release of the incremental-movement user interface **115** (e.g., depressing and releasing a button with a finger, depressing and releasing a paddle with a finger, touching and untouching an icon on a touchscreen) corresponds to one discrete input. In examples in which more than one discrete input is entered through the incremental-movement user interface **115** in a time period, the discrete inputs are individual and distinct from each other. For example, one finger depression and subsequent finger release is one discrete input and another finger depression and finger release in the time period is a second discrete input.

[0045] The operation of the incremental-movement user interface **115** under certain conditions, e.g., operation according to method **400** in FIG. 4, moves the vehicle **100** an incremental distance. The incremental distance is a relatively small distance that initiates when the vehicle **100** is at a standstill and ends when the vehicle **100** is at a standstill. As set forth above, the vehicle **100** is at a standstill when all wheels **145** of the vehicle **100**, i.e., all four wheels **145** in four-wheeled vehicle examples, are stationary relative to the body of the vehicle **100** and the ground. The incremental distance, as an example, may be between 0.5 inches and 1.5 inches (12.7 mm-38.1 mm) depending on the number of discrete inputs to the incremental-movement user interface **115** in the time period. Examples of the incremental distance traveled by the vehicle **100** by operation of the incremental-movement system **160** are shown in FIGS. 1A and 1B. The incremental distance in FIGS. 1A and 1B is identified with reference “d.” The magnitude of the incremental distance in FIGS. 1A and 1B are shown by way of example and, in some examples, may be exaggerated for illustrative purposes. An incremental distance traveled by the vehicle by operation is also shown, but not identified with “d”, by operation of the incremental-movement system **160** in FIG. 3.

[0046] The input from the incremental-movement user interface **115** may map onto a torque outputted by the propulsion system **150** over time, i.e., $t=f(I)$, in which t is the torque outputted by the propulsion system **150** and/is the input. In such an example, the systems of the vehicle **100** operate to provide torque to one or more traction motors **110** over a time to target movement of the vehicle **100** in the incremental distance. For example, the torque outputted by the propulsion system **150** targets movement of the vehicle **100** in the incremental distance based on input to the incremental-movement user interface **115** during the time period.

[0047] The input from the incremental-movement user interface **115** may map onto a distance traveled by the vehicle **100** before stopping, i.e., $d=f(I)$, in which d is the distance traveled by the

vehicle **100** and/is the input. In such an example, the systems of the vehicle **100** operate rotate at least one of the traction motors **110** to a rotational position to the target movement of the vehicle **100** in the incremental distance, i.e., d is the incremental distance. For example, the vehicle computer **105** may control acceleration and braking of the vehicle **100** to travel the incremental distance and then stop movement of the vehicle **100**. To travel the incremental distance, the vehicle computer **105** may actuate the propulsion system **150** until the vehicle **100** has traveled the incremental distance and actuate the brake system **120** to brake the vehicle **100** to a stop upon the vehicle **100** traveling the incremental distance.

[0048] The input from the incremental-movement user interface **115** may map onto a speed at which the vehicle **100** travels, i.e., $v=f(I)$, in which v is the vehicle velocity and/is the input. In such an example, the systems of the vehicle **100** operate rotate at least one of the traction motors **110** at a velocity to the target movement of the vehicle **100** in the incremental distance. For example, the vehicle computer **105** may control acceleration and braking of the vehicle **100** to reach a velocity to travel the incremental distance and then stop movement of the vehicle **100**.

[0049] The human driver may use the incremental-movement user interface **115** when the vehicle **100** is at a standstill to selectively move the vehicle **100** incremental distances. As set forth above, the incremental distances are relatively small. As examples, the human driver may bring the vehicle **100** to a standstill and selectively use the movement in incremental distances in situations such as to parallel park the vehicle **100** (e.g., see FIG. 2), to align a hitch of the vehicle **100** with a trailer to be towed, to driver around or on top of a ground feature such as a rock or stump during off-road use, or any other instance in which incremental movement from a standstill is useful to the human driver. When the human driver brings the vehicle **100** to a standstill, the human driver can identify a desired position of the vehicle **100** and provide discrete inputs to the incremental-movement user interface **115** to incrementally move the vehicle **100** to the desired position. If a longer incremental distance is desired, the vehicle driver may provide multiple discrete inputs to the incremental-movement user interface **115** in a time period to increase the next incremental distance under the conditions described herein. Movement of the vehicle **100** in multiple, separate, incremental distances may be controlled by the human driver by discrete inputs in different time periods. In such instances, the incremental movements can be consecutively applied by the human driver and/or can be interspersed with control of the movement of the vehicle **100** by other ways of controlling the propulsion system **150** of the vehicle **100**, e.g. control with an accelerator pedal and/or brake pedal. In the event the incremental movements are interspersed with use other controls of the propulsion system **150**, e.g. the accelerator pedal and/or brake pedal, the vehicle **100** is brought to a standstill before the next incremental movement can be made with the use of the incremental-movement user interface **115**. The incremental movement of the vehicle **100** may be forward movement of the vehicle **100** or rearward movement of the vehicle **100**, depending on the gear selection of the vehicle **100**. In other words, the incremental distance may be a distance in a vehicle-forward direction or a distance in a vehicle-rearward direction based other operating conditions of the vehicle **100**.

[0050] As set forth above, the vehicle driver may provide multiple discrete inputs to the incremental-movement user interface **115** in a time period to increase the next length of the incremental distance under the conditions described herein. The time period is a relatively short period, e.g., three seconds. As an example, if conditions are met, including that the vehicle **100** is at a standstill, a discrete input to the incremental-movement user interface **115** initiates the operation of the incremental-movement system **160**, e.g., operation of method **400**. The time period begins upon one discrete input to the incremental-movement user interface **115**. For example, the vehicle computer **105** may receive the first discrete input that initiated the operation of the incremental-movement system **160** and may wait for the time period to detect any additional discrete inputs. After the expiration of the time period, the vehicle computer **105** calculates the length of the incremental distance based on the number of discrete inputs entered, i.e., the discrete input to

initiate the time period and any additional discrete inputs that may have been entered during the time period. Thereafter, the operation of method **400** continues. The time period may be a constant, i.e., the same length of time each time the incremental-movement system **160** is operated. In such an example, the time period is predetermined and programmed into the vehicle computer **105** as a constant.

[0051] The incremental-movement system **160** may limit the number of discrete inputs in any one time period. In some examples, the number of discrete inputs may be limited to a total of three discrete inputs to initiate the time period and entered during the time period. In the event more the number of discrete inputs during one time period exceeds the limit, the vehicle computer **105** ignores the discrete inputs above the limit. After completion of the movement in the incremental distance and return of the vehicle **100** to standstill, a new batch of one or more discrete inputs may be entered if all other conditions are met to again perform method **400** for another incremental movement.

[0052] The vehicle computer **105** is programmed to operate the other components of the incremental-movement system **160** as described herein. Specifically, the vehicle computer **105** is programmed to perform the method **400** shown in FIG. 4 and as described herein. Use of “in response to,” “based on,” and “upon determining” herein indicates a causal relationship, not merely a temporal relationship.

[0053] With reference to FIG. 4, the method **400** includes receiving at least one discrete input from the incremental-movement user interface **115**. Upon entry of one discrete input by the human driver, e.g., depressing or touching the incremental-movement user interface **115** as described above, the method **400** is initiated. As described above, the time period is initiated at the entry of the first discrete input by the human driver for the current request for operation of the incremental-movement system **160**.

[0054] As shown in block **410**, the vehicle **100** being at a standstill is a condition for operation of the method **400**. If the vehicle **100** is not at a standstill when the first discrete input is detected, the current request for operation of the incremental-movement system **160** is aborted and the method **400** returns to block **405** to restart the method **400**. An indicator, e.g., a visual or audible indicator, may be provided to the human driver indicating that the requested operation of the incremental-movement system **160** was not carried out because the vehicle **100** was not at a standstill.

[0055] In some examples, a condition to initiate the operation of method **400** includes that a brake is not manually applied by the human driver to the vehicle **100**, e.g., a brake pedal is not depressed by the human driver in examples including a brake pedal. In some examples, a condition to initiate the operation of the method **400** includes detection that a steering angle of the steering wheel **145** is outside of a predetermined range, e.g., based on a measurement by the steering-wheel position sensor **130**. Such an example may be used in operation of the method **400** for parallel parking in which the initial angle of the steering wheel **145** must be turned relatively sharply (e.g., see FIG. 2). In such examples, the predetermined range may be, for example, ± 30 degrees from center, i.e., the condition is satisfied when the steering wheel **145** is turned more than $+30$ degrees from center or more than -30 degrees from center. In such examples, the method **400** includes determining that a steering angle of a steering wheel **145** of the vehicle **100** is outside of a predetermined range and then moving the vehicle **100** the incremental distance based on the determination that the steering angle is outside of the predetermined range in addition to the other conditions described herein.

[0056] With reference to block **415**, the method **400** includes calculating the length of the incremental distance that will be targeted for vehicle movement. As set forth above, the entry of the first discrete input by the human driver in the current request for operation of the incremental-movement system **160** initiates the time period. Additional discrete inputs by the human driver during the time period increase the length of the incremental distance that will be targeted for vehicle movement. In other words, the human driver can select an increase in the incremental

distance by entering multiple discrete inputs to the incremental-movement user interface **115**. As set forth above, the number of discrete inputs that can be entered during the time period may be limited.

[0057] In response to the at least one discrete input during the time period, the method **400** includes moving the vehicle **100** an incremental distance, as shown in block **420**. After the length of the incremental distance is calculated, the method includes commanding a traction motor **110** to move the vehicle **100** in the incremental distance. The number of discrete inputs during the time period and the corresponding length of the incremental distance is used in determining a motor input for at least one traction motor **110**. Block **420** includes commanding at least one traction motor **110** to rotate based on that motor input.

[0058] The calculation of the length of the incremental distance based on additional discrete inputs during the time period and the associated movement of the vehicle **100** may be made in any suitable way. Several examples are described herein merely by way of example. In any of the examples, the propulsion system **150** and the braking system may be used interactively to target movement of the vehicle **100** in the incremental distance.

[0059] As one example of the calculation of the length of the incremental distance and associated movement of the vehicle **100**, the method **400** may include accessing a lookup table to identify a length of the incremental distance based on the number of discrete inputs. Such an example is an open-loop system. As one example, the lookup table may include vehicle velocity over time. One such example is shown in FIGS. 5A and 5B. In FIG. 5A, the top graph shows a single discrete input during the time period. In the middle graph of FIG. 5A, the solid line depicts the ramp up and ramp down of vehicle velocity over time based on the lookup table to target movement in the incremental distance. In the middle graph of FIG. 5A, the dotted lines shows a hypothetical example of the actual velocity of the vehicle **100** under the operation of the method **400**. The bottom graph of FIG. 5A shows a hypothetical example of the distance traveled by the vehicle **100** under the application of the velocity in the middle graph of FIG. 5A. In FIG. 5B, the top graph shows the entry of three discrete inputs during the time period. In the middle graph of FIG. 5B, the solid line depicts the ramp up, maintenance, and ramp down of vehicle velocity over time based on the lookup table to target movement in the incremental distance. In the middle graph of FIG. 5B, the dotted lines shows a hypothetical example of the actual velocity of the vehicle **100** under the operation of the method **400**. The bottom graph of FIG. 5B shows a hypothetical example of the distance traveled by the vehicle **100** under the application of the velocity in the middle graph of FIG. 5B. The peak velocity in the middle graphs of FIGS. 5A and 5B are the same and may be, as an example, 0.1 kph. As another example, the lookup table may include, for example, a torque profile for each number of discrete inputs during the time period. The torque profile includes the variation in torque over time. The torque is applied by the traction motor **110** to the wheel **145** according to the torque profile. The distance traveled is not actively measured and assumed to be achieved with a certain vehicle velocity or torque profile over time.

[0060] As another example of the calculation of the length of the incremental distance and associated movement of the vehicle **100**, the method **400** may be an closed loop system that receives measurement of the distance traveled and returns the vehicle **100** to a standstill at incremental distance. For example, the method **400** includes commanding at least one of the traction motors **110** to initiate movement of the vehicle **100** in the incremental step from standstill, measuring travel of the wheel **145** powered by the traction motor **110** from standstill, and returning the vehicle **100** to standstill based on measurement of the travel of the wheel **145**. The measurement of the distance traveled by the vehicle **100** may be, for example, taken by the wheel position sensor **135** and/or the resolver **140**, as described above. An example is shown in FIG. 6 in which the top graph, as an example, shows three discrete inputs during the time period. In the middle graph of FIG. 6, the solid line depicts the ramp up and ramp down of vehicle velocity over time based on the measured travel distance to target movement in the incremental distance. In the

middle graph of FIG. 6, the dotted lines shows a hypothetical example of the actual velocity of the vehicle **100** under the operation of the method **400**. The bottom graph of FIG. 6 shows a hypothetical example of the distance traveled by the vehicle **100** under the application of the velocity in the middle graph of FIG. 6. In such examples, the peak velocity (e.g., the peak velocity in the middle graph of FIG. 6) may be constant for each operation of the method **400** regardless of the number of discrete inputs. As an example, the peak velocity may be 0.1 kph.

[0061] As another example of the calculation of the length of the incremental distance and associated movement of the vehicle **100**, the method **400** may include steps to reduce or remove gear lash as a variable to the distance traveled by the vehicle **100**. Gear lash, as is known, is the play, or slop, between two gears that can allow relative movement between two gears at initial rotation. In the example shown in FIG. 7, the method includes staggering power between two traction motors **110** to reduce or eliminate gear lash from target movement of the vehicle **100** along the incremental distance. Specifically, the method **400** may include commanding a first traction motor **110** to rotate a first wheel **145** of the vehicle **100** while measuring traveled distance of a second wheel **145** of the vehicle **100**. In such an example, the first wheel **145** may be one of a front wheel **145** or a rear wheel **145** and the second wheel **145** is the other of the front wheel **145** or a rear wheel **145**. In such an example, the propulsion system **110** includes at least two traction motors **110**, i.e., one for the first wheel **145** and one for the second wheel **145**. For example, the propulsion system **110** in such examples may include one traction motor **110** that drives one or both rear wheels **145** and another traction motor **110** that drives one or both of the front wheels **145**. As the first traction motor **110** moves the vehicle **100**, any gear lash in the gearing to the second wheel **145** is avoided, i.e., the play between gears is avoided as second wheel **145** is rotated by movement of the vehicle **100** as powered by the first traction motor **110**. After powering the first traction motor **110**, the method **400** then includes identifying occurrence of gear lash in the rotation of the first wheel **145** (since it is driving the vehicle). As an example, a lash crossing can be detected with the measurement of the movement of the first traction motor **110**. The lash crossing may be, for example, a detected spike in the measured motor speed. After identifying the occurrence of gear lash in the rotation of the first wheel **145**, the method then includes commanding a second traction motor **110** that powers the second wheel **145** to rotate the second wheel **145** while measuring the traveled distance of the first wheel **145**. The measurement of the traveled distance of the second wheel **145** and the subsequent measurement of traveled distance of the first wheel **145** can be used, in combination, to measure the traveled distance of the vehicle **100**. The elimination of gear lash increases the accuracy of the measurement of the traveled distance. The measurement of the traveled distance of the first wheel **145** and the second wheel **145** described in this example may be taken by, for example, the wheel position sensor **135** and/or the resolver **140** of that wheel **145**.

[0062] In the example in FIG. 7, which the top graph, as an example, shows three discrete inputs during the time period. In the middle graph of FIG. 7, the solid line depicts the ramp up and ramp down of vehicle velocity over time based on the measured travel distance to target movement in the incremental distance. In the middle graph of FIG. 7, the dotted lines shows a hypothetical example of the actual velocity of the vehicle **100** under the operation of the method **400**. The lash crossing detected in the second wheel **145** is identified with the spike in the dotted line. The spike is exaggerated to illustrate the lash crossing for this description. In the bottom graph of FIG. 7, the solid line is the targeted distance traveled over time under the application of the velocity in the middle graph of FIG. 7. The dotted line in FIG. 7 shows a hypothetical example of the actual distance traveled by the vehicle **100** under the application of the velocity in the middle graph of FIG. 7. In such examples, the peak velocity (e.g., the peak velocity in the middle graph of FIG. 7) may be constant for each operation of the method **400** regardless of the number of discrete inputs. As an example, the peak velocity may be 0.1 kph.

[0063] The method **400** includes returning the vehicle **100** to a standstill at the end the incremental movement. The return to standstill may be caused by reducing power to the traction motors **110**

and/or braking one or more wheels **145** of the vehicle **100** during movement of the vehicle **100** in the incremental distance to return the vehicle **100** to a standstill. After the vehicle **100** is returned to standstill, the method **400** returns to block **405**. At that point, the method **400** can be repeated for another incremental movement of the vehicle **100**. As other options, after returning to standstill, the vehicle **100** may be moved in other ways, e.g., with operation of the accelerator pedal.

[0064] In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Ford Sync® application, AppLink/Smart Device Link middleware, the Microsoft Automotive® operating system, the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the Mac OSX and iOS operating systems distributed by Apple Inc. of Cupertino, California, the BlackBerry OS distributed by Blackberry, Ltd. of Waterloo, Canada, and the Android operating system developed by Google, Inc. and the Open Handset Alliance, or the QNX® CAR Platform for Infotainment offered by QNX Software Systems. Examples of computing devices include, without limitation, an on-board vehicle computer **105**, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

[0065] Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Matlab, Simulink, Stateflow, Visual Basic, Java Script, Python, Perl, HTML, etc. Some of these applications may be compiled and executed on a virtual machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random access memory, etc.

[0066] A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless communication, including the internals that comprise a system bus coupled to a processor of a computer. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

[0067] Databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), a nonrelational database (NoSQL), a graph database (GDB), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above, and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system, and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

[0068] In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

[0069] The adjectives “first” and “second” are used throughout this document as identifiers and are not intended to signify importance, order, or quantity. Use of “in response to” and “upon determining” indicates a causal relationship, not merely a temporal relationship.

[0070] The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. A method comprising: receiving at least one discrete input within a predetermined time period from a human-machine interface while the vehicle is at a standstill; and in response to the at least one discrete input, moving the vehicle an incremental distance and returning the vehicle to a standstill.
2. The method as set forth in claim 1, further comprising determining that a steering angle of a steering wheel of the vehicle is outside of a predetermined range; and wherein moving the vehicle the incremental distance is based on the determination that the steering angle is outside of the predetermined range.
3. The method as set forth in claim 1, further comprising commanding a traction motor to move the vehicle in the incremental distance.
4. The method as set forth in claim 3, further comprising braking a wheel of the vehicle during movement of the vehicle in the incremental distance to return the vehicle to a standstill.
5. The method as set forth in claim 1, further comprising braking a wheel of the vehicle during movement of the vehicle in the incremental distance to return the vehicle to a standstill.
6. The method as set forth in claim 1, further comprising: receiving more than one discrete input within the predetermined time period; and calculating the length of the incremental distance based on the number of discrete inputs.
7. The method as set forth in claim 6, further comprising: determining a motor input for at least one traction motor based on the length of the incremental distance; and commanding the at least one traction motor to rotate based on the motor input.
8. The method as set forth in claim 1, further comprising receiving more than one discrete input within the predetermined time period and accessing a lookup table to identify a length of the incremental distance based on the number of discrete inputs.
9. The method as set forth in claim 1, further comprising: commanding a first traction motor to rotate a first wheel of the vehicle while measuring traveled distance of a second wheel of the vehicle; then, identifying occurrence of gear lash in the rotation of the first wheel; and then, commanding a second traction motor to rotate the second wheel while measuring traveled distance of the first wheel.
10. The method as set forth in claim 1, further comprising: commanding a traction motor to initiate movement in the incremental step from standstill; measuring traveled distance of a wheel of the vehicle from standstill; and returning the vehicle to standstill based on measurement of the traveled distance of the wheel.
11. A system comprising: a computer having a processor and a memory storing instructions executable by the processor to: receive at least one discrete input within a predetermined time

period from a human-machine interface while the vehicle is at a standstill; and in response to the at least one discrete input, move the vehicle an incremental distance and returning the vehicle to a standstill.

12. The system as set forth in claim 11, wherein the memory stores instructions executable by the processor to: receive an indication that a steering angle of a steering wheel is outside of a predetermined range; and move the vehicle the incremental distance based on the indication that the steering angle is outside of the predetermined range.

13. The system as set forth in claim 11, wherein the memory stores instructions executable by the processor to command a traction motor to move the vehicle in the incremental distance.

14. The system as set forth in claim 13, wherein the memory stores instructions executable by the processor to brake a wheel of the vehicle during movement of the vehicle in the incremental distance to return the vehicle to a standstill.

15. The system as set forth in claim 11, wherein the memory stores instructions executable by the processor to brake a wheel of the vehicle during movement of the vehicle in the incremental distance to return the vehicle to a standstill.

16. The system as set forth in claim 11, wherein the memory stores instructions executable by the processor to: receive more than one discrete input within the predetermined time period; and calculate the length of the incremental distance based on the number of discrete inputs.

17. The system as set forth in claim 16, wherein the memory stores instructions executable by the processor to: determine a motor input for at least one traction motor based on the length of the incremental distance; and command the at least one traction motor to rotate based on the motor input.

18. The system as set forth in claim 11, wherein the memory stores instructions executable by the processor to: receive more than one discrete input within the predetermined time period; and access a lookup table to identify a length of the incremental distance based on the number of discrete inputs.

19. The system as set forth in claim 11, wherein the memory stores instructions executable by the processor to: command a first traction motor to rotate a first wheel while measuring traveled distance of a second wheel; then, identify occurrence of lash in the rotation of the second wheel; and then, command a second traction motor to rotate the second wheel while measuring traveled distance of the first wheel.

20. The system as set forth in claim 11, wherein the memory stores instructions executable by the processor to: command the traction motor to initiate movement in the incremental step from standstill; measure traveled distance of the wheel of the vehicle from standstill; and return the vehicle to standstill based on measurement of the traveled distance of the wheel.
